Optimization: Critical Points & Lagrange Multipliers

Section 1: Finding Critical Points

1 What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$, meaning both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

2 Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$

b. $(-2, 3)$

c. $(4, -6)$

d. $(0, 0)$

3 Find all critical points of $g(x, y) = x^3 - 3x + y^2$.

a. $(1, 0)$ and $(-1, 0)$

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

Section 2: The Second Partials Test (Hessian Discriminant)

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4 What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

5 If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f has a local minimum at the point

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

6 If at a critical point $D < 0$, what does the test conclude?

a. f has a saddle point

b. f has a local maximum

c. f has a local minimum

d. The test is inconclusive

Section 3: Classifying Extrema and Inconclusive Cases

7 Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. Saddle point

b. Local minimum

c. Local maximum

d. Inconclusive

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8 What does the Second Partials Test conclude when $D = 0$?

a. The test is inconclusive; higher-order analysis or algebraic examination is needed

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

9 Classify $(2, -3)$ for $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. Local minimum

b. Local maximum

c. Saddle point

d. Inconclusive

Section 4: Lagrange Multipliers for Constrained Optimization

10 What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?

a. $\nabla f = \lambda \nabla g$ (and $g(x, y) = k$)

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

11 Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. Maximum is 25 at $(5, 5)$

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

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12 What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant k ($\frac{df^*}{dk}$)

b. The curvature of the constraint surface

c. The distance from the origin to the optimal point

d. The second derivative along the constraint

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